

Air Quality Intelligence System

Advanced Data Warehouse Technologies — Course Report

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1 Team Information

Field	Value
Student	Péter Dobosi
Neptun code	MW79ON
Course	Advanced Data Warehouse Technologies
Semester	2025/26 Spring
Repository	github.com/dobosipeter/advanced-...-submission
Live system	https://mw79on-demo.online
API docs	https://api.mw79on-demo.online/docs

2 System Architecture

The system implements an end-to-end data warehouse pipeline: ingesting real-time air quality data from the OpenAQ public API, staging it in raw JSON form, loading it into a normalized OLTP schema, then transforming it into an analytical star schema in the `dw` schema. A machine learning model provides PM2.5 predictions stored back into the warehouse.

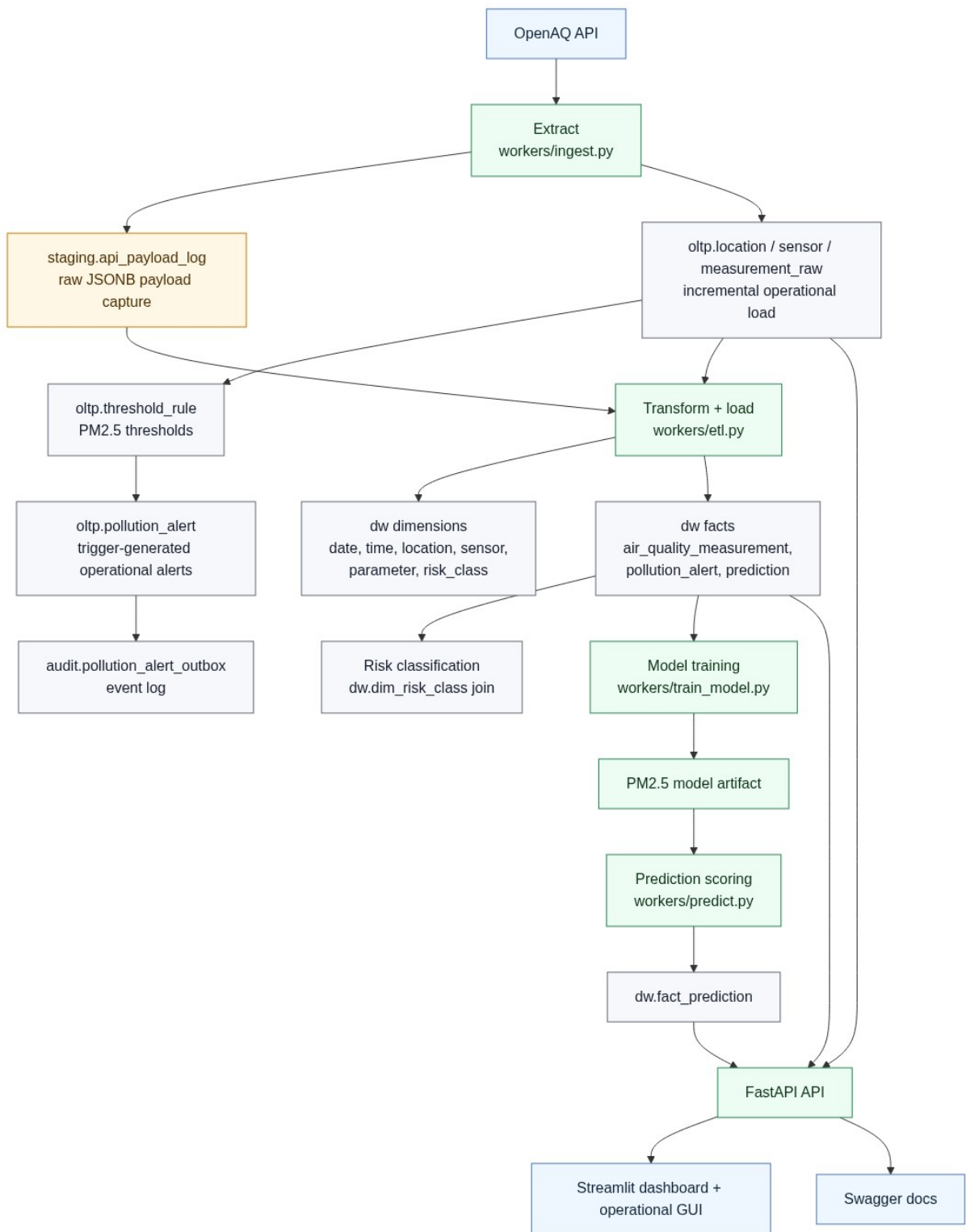


Figure 1: End-to-end data flow: source → staging → OLTP → ETL → Data Warehouse → dashboard.

3 Course Technologies Demonstrated

The following topics from the Advanced Data Warehouse Technologies course are actively demonstrated:

3.1 Star Schema Design (Chapter 1)

The DW uses a **fact constellation** (galaxy schema): three fact tables sharing conformed dimensions. This is a standard DW pattern when multiple business processes (measurements, alerts, predictions) share the same dimensional context (date, time, location, sensor, parameter).

- **Conformed dimensions:** `dim_date` and `dim_time` are pre-populated and shared across all facts.
- **Degenerate dimensions:** measurement unit stored directly on the fact row.
- **Hierarchies:** date dimension supports year → quarter → month → day drill-down.
- **Reference dimension:** `dim_risk_class` maps continuous PM2.5 values to categorical severity bands.

3.2 Slowly Changing Dimensions — SCD Type 2 (Chapter 2)

- `dim_location` and `dim_sensor` implement **SCD Type 2** with columns: `valid_from`, `valid_to`, `is_current`.
- Change detection uses **MD5 hash comparison** of attribute values.
- When a location changes (e.g., name update, coordinate correction): the current row is expired (`valid_to = now`, `is_current = false`) and a new row is inserted with a new surrogate key.
- Historical queries can join facts to the dimension version that was valid at measurement time.

3.3 ETL Automation (Chapter 3)

- `workers/etl.py` implements a complete ETL pipeline in Python (analogous to SSIS dataflows):
 1. **Extract:** watermark-based incremental pull from OLTP (only rows since last ETL run).
 2. **Transform:** SCD2 hash comparison, date/time key derivation, risk classification lookup.
 3. **Load:** bulk insert into fact tables; duplicate-safe via `ON CONFLICT DO NOTHING` (idempotent re-runs).
- Pipeline idempotency: re-running ETL on the same data produces no duplicates.

3.4 Automated Refresh & Scheduling (Chapter 3 – SSIS Jobs)

The pipeline is scheduled via system cron on the production VM:

- **Every 3 hours:** `run_pipeline.sh ingest-etl` — fetch new data and load into DW.
- **Daily at 06:00 UTC:** `run_pipeline.sh predict-only` — generate next-period predictions.
- **Weekly Sunday 03:00 UTC:** `run_pipeline.sh train-predict` — retrain ML model and predict.

3.5 Reports, KPIs & Visualization (Chapter 5 – PowerBI/Dashboards)

The Streamlit dashboard provides analytical views from the DW:

- **Air Quality Overview** — daily aggregated trend charts by parameter.
- **High-Risk Periods** — heatmaps showing hours/days with elevated pollution.
- **Prediction Insights** — predicted vs. actual comparison with error metrics.
- **KPI cards** on the home page: total measurements, active alerts, prediction accuracy.

3.6 Machine Learning Integration (100% criterion)

- A **Random Forest** model is trained on historical PM2.5 data from the DW.
- Features: current value, lag-1, rolling 3h/6h means, temporal (hour, weekday, month), sensor/location keys.
- Predictions are stored as **fact_prediction** with associated **risk_class_key**.
- Model metrics tracked: MAE, RMSE, R^2 (currently $R^2 \approx 0.73$).

3.7 DB Administration & Alerts (Chapter 7)

- Threshold-based alerting rules trigger pollution alerts automatically.
- Alert lifecycle tracked in DW as **fact_pollution_alert**.
- Automated backups (daily **pg_dump**) and scheduled maintenance (**VACUUM/ANALYZE**).

4 Functional Specification

1. **Data Ingestion** — Incremental and bulk ingest from OpenAQ v3 API with error tracking.
2. **ETL Pipeline** — Extracts from OLTP, applies SCD2, loads facts incrementally.
3. **Star Schema (Fact Constellation)** — Three fact tables with six shared dimensions.
4. **SCD Type 2** — History-preserving dimensions with hash-based change detection.
5. **Risk Classification** — PM2.5 values classified into Low / Moderate / High / Critical.
6. **PM2.5 Prediction** — Random Forest model; predictions stored in **fact_prediction**.
7. **Automated Refresh** — Cron-scheduled pipeline (every 3h ingest-ETL, daily predict).
8. **Analytical Dashboard** — Streamlit with trend charts, heatmaps, KPIs.
9. **Threshold Alerts** — Configurable rules; alerts loaded into DW as facts.
10. **Deployment** — Docker Compose on Hetzner Cloud with automatic HTTPS.

5 Logical Data Model (Star Schema)

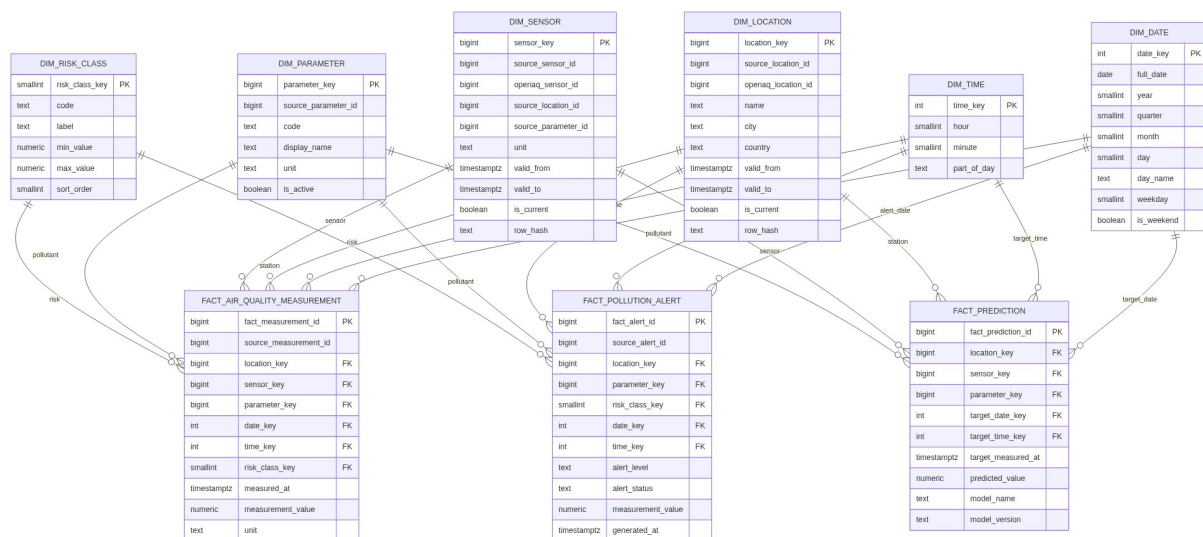


Figure 2: Data warehouse star schema (fact constellation with shared dimensions).

Fact Tables

The design uses a **fact constellation** (also called galaxy schema), where multiple fact tables share conformed dimensions. This is standard practice when a DW serves multiple related business processes.

Table	Grain	Measures
fact_air_quality_measurement	One reading per sensor per timestamp	measurement_value, unit
fact_pollution_alert	One alert event	measurement_value, level, status
fact_prediction	One prediction per location/time	predicted_value, model info

Table 1: Fact tables — three business processes sharing conformed dimensions.

Dimension Tables

Table	Type	Description
dim_date	Conformed, pre-populated	Calendar attributes (year, quarter, month, weekday, is_weekend)
dim_time	Conformed, pre-populated	Hour/minute with part-of-day classification
dim_parameter	Type 1	Pollutant definitions (pm25, pm10, no2, o3)
dim_location	SCD Type 2	Monitoring stations with validity range
dim_sensor	SCD Type 2	Sensors with validity range
dim_risk_class	Reference	PM2.5 concentration bands (WHO guidelines)

Table 2: Dimension tables.

6 Technologies Used

Technology	Role
PostgreSQL 16	OLTP + Data Warehouse (schema separation)
Python 3.12	ETL scripts, ML training, prediction, ingestion
scikit-learn	Random Forest model for PM2.5 prediction
pandas	Data transformation and feature engineering
Streamlit	Analytical multi-page dashboard
Plotly	Interactive charts (line, bar, heatmap)
FastAPI	REST API serving DW data
Docker Compose	Service orchestration (5 containers)
Caddy 2	HTTPS reverse proxy (Let's Encrypt)
GitHub Actions	CI/CD deployment pipeline
Hetzner Cloud	Production VM (Ubuntu 24.04)